

Cognitive Wellness Assessment: A Comprehensive Review of Deep Learning and Machine Learning Approaches for Mental Stress Detection in the Digital Age

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Abstract—Mental stress is a big problem nowadays, particularly in young people. The age group that was formerly seen as the most carefree is presently experiencing a great deal of stress. Increased stress in today's world causes numerous issues like depression, suicide, heart attacks, and strokes. Stress identification is important because it can help to prevent chronic stress and its negative health consequences, improve productivity and performance, and reduce the risk of accidents and injuries. It can also increase self-awareness and improve communication and relationships. Effective stress management is essential for preserving general wellbeing and guarding against the harmful consequences of long-term stress, which can have an adverse influence on relationships, mental and physical health, and physical health. By incorporating stress management techniques into daily life, individuals can effectively reduce stress levels, enhance their quality of life, and promote resilience. Specializing in deep learning and machine learning methods, this paper offers an extensive overview of the research on stress identification. Their development over a number of years is covered. Furthermore, the study makes recommendations for future work, with a focus on increasing accuracy and utilizing new technologies. For scholars and practitioners in the field, this paper provides insightful information about the prospective applications of deep learning and machine learning in stress identification.

Keywords- Mental health, deep learning, machine learning, stress identification

I. INTRODUCTION

Stress is an issue that people in today's society face, affecting their personal well-being, careers, and health, among other aspects of their existence. Since unchecked stress can result in a variety of physical and mental health issues, mastering the art of stress detection and management is crucial. New avenues for stress identification have been opened up in recent years by a variety of deep learning and machine learning techniques, providing creative approaches to identify and manage stress. More than 70% of people are under stress [1]. An immune system that is compromised by prolonged stress is susceptible to heart disease [3,4], cancer [4], diabetes [5, 6], depression [7] and drug addiction [8]. Stress poses significant risks to both mental and physical,

necessitating the development of efficient stress identification techniques. The ability to continuously detect stress could empower individuals to manage daily activities to mitigate its impact, while aiding medical professionals in treating stress-related disorders. This paper explores the pivotal role of deep learning and machine learning in stress identification, investigating their applications across diverse domains. The review encompasses a comprehensive examination of relevant literature, presenting key findings, insights, and addressing research gaps and challenges in implementing these technologies. In order to be a useful tool for academics, professionals, and decision-makers, it seeks to shed light on the growing importance of deep learning and machine learning in stress identification and management. The assessment will include research from 2010 to 2023 that examine the use of deep learning (DL) and machine learning (ML) in a variety of contexts, such as healthcare, workplaces, and applications related to personal well-being. Stress identification and management require more objective methods than conventional self-reported measures.

II. LITERATURE REVIEW

The proposed review will encompass peer-reviewed research articles, conference papers, and other scholarly publications on stress identification in humans, covering the period from 2010 to 2023. Included studies will focus on the use of DL and ML for stress identification, utilizing various data sources. The review will examine studies conducted in diverse settings, including healthcare, workplaces and personal well-being applications. The field of stress detection using DL and ML is rapidly expanding and provides new insights into the identification and tracking of stress thanks to a substantial body of research. This review draws from a wide range of sources and studies in the literature to offer a thorough analysis of major findings, methodology, and issues. Using DL and ML methods is one of the significant advances in stress identification. Many previous techniques had looked at a range of physiological signals, including the ECG and electrodermal activity

data. Using a variety of approaches, the review has been divided into multiple sections.

A. Using electromyography

Using electromyography sensors, Setz et al. [9] adopted a traditional ML approach to classify emotions and detect stress by analyzing physiological signals using algorithms such as random forest, K-nearest neighbor (KNN), decision tree, support vector machine (SVM), linear discriminant analysis (LDA) and others.

Lopez et al. [10] aimed to detect stress in drivers by analyzing electromyographical (EMG) signals, utilizing statistical time features (STFs) and the Kruskal–Wallis–Mann–Whitney (KWM) method. Their approach, particularly employing Support Vector Machine (SVM) with selected STFs, achieved a significant 96% accuracy in real-time stress identification, surpassing methods using multiple physiological signals. While the study demonstrated simplicity without preprocessing algorithms, it acknowledged the need for further validation with a larger database, potential recalibration, and the development of portable, non-intrusive physiological monitoring systems. Overall, the research presents a promising method for efficient stress identification in drivers, contributing to improved road safety.

B. Using ECG Signals

Wearable sensors were also utilized by Grundlehner et al. [11] to assess physiological signals in order to diagnose mental stress. 19 distinct physiological traits were calculated from the patients' recorded ECG, respiration, conductance of skin and EMG. After correlations were looked at and feature values were normalized, nine of these 19 features were selected for further research. After that, PCA was utilized to cut the sample of nine features down to seven. These factors, when combined with Fisher's Least Square Linear Classifier, Quadratic Bayes Normal Classifier and KNN Classifier, were able to accurately distinguish an individual's stress from non-stress settings with an accuracy of 80%.

Keshan et al. [12] used ECG data to look for signs of stress. 88.24% accuracy in identifying three stress levels (low, medium, and high) was attained by the study using ML algorithms solely on ECG signals. By employing the Naive Bayes algorithm on the variation in average heart rate characteristic, it was able to identify high levels of stress with 100% accuracy when compared to a person's rest time. Decision trees (J48) on all eight features were used, and 98% accuracy was attained in identifying high stress levels. Skin conductance, respiration rate, and blood pressure are examples of signals that were not included in the study. This study showed promising results in using ECG monitoring to detect stress levels. To integrate other signals in the study and validate the results on a larger sample size, more research is necessary.

Mashuk et al. [13] used the ECG characteristic in addition to other available biological signals to categorize stress

conditions. Because feature extraction techniques for ECG are available and there are several mobile medical-grade recording devices that make ECG recording easily accessible, ECG was chosen as the primary option. Since it can identify respiratory signal information via the EDR method (i.e., respiration derived from ECG), ECG is chosen since it eliminates the need for a separate respiration measurement sensor system. They achieved about 98.6% accuracy using the SVM classification algorithm with RR interval, QT interval, and EDR characteristics. This result, however, is inadequate since the research only considered one signal—the ECG—instead of other essential body signals that are also important for triggering stress.

Giannakakis et al. [14] aimed to identify reliable heart rate variability (HRV) parameters for stress recognition. The experiment involved diverse stressors, and personalized baselines were established to eliminate inter-subject variability. HRV features from ECG signals were transformed using pairwise transformation, and the mRMR algorithm identified 11 top-ranked features for stress anticipation. With a 10-fold cross-validation strategy, the suggested system's SVM classification accuracy was 84.4%. Despite acknowledging the potential benefits of multimodal information, the focus on ECG signals alone demonstrated efficiency, especially for wearable devices. Limitations included the need for a larger sample size and potential task-related variations in stress response. The study concluded that using only ECG recordings could efficiently contribute to stress identification.

A DL system for identifying stress in real time using ECG signals was suggested by Zhang et al. [15]. Convolutional neural networks (CNNs) and the bidirectional long short-term memory (BiLSTM) served as the foundation for the framework. CNNs were able to learn discriminative features from raw ECG data, and BiLSTM networks were able to learn long-range temporal dependencies in the data. Using a 10-second ECG signal, the suggested framework attained an accuracy of 0.865 on three stress levels, which is much higher than the accuracy of existing approaches (0.637). The proposed framework was also able to achieve high accuracy in real-time, with an accuracy of 0.837 on a 5-second ECG signal. Personality differences among experimenters can have a substantial effect on stress levels, but were not considered in this investigation. The study was conducted on a relatively small sample size (34 subjects).

C. Using HyperSpectral Imaging

With an accuracy of 88.1%, Chen et al. [16] used a binary classifier on hyperspectral photos of people's faces. The reported classification findings were reliant on the individual and differ throughout the participant groups due to the small number of participants. However, if big data sets are used, the classification threshold ought to get closer to a steady constant.

Using hyperspectral imaging, Nagaraj et al. [17] conducted a biometric study under stress. A significant change in histogram density of pixel intensities in hyperspectral images was observed in spectral band 13, corresponding to

a near-infrared wavelength, when comparing hyperspectral images taken 1 second before and 1 second after a paintball hit. This change was consistent across six subjects and suggests an association with stress or psychological states related to the painful event. A density parameter quantified this change, with mean values of 3.13 before the shot and 5.92 after the shot. It's unclear whether the study's findings would apply to other kinds of stressors because it only examined one kind of stressor—a paintball strike. It would be beneficial to study a greater range of stressors. The study only used hyperspectral data from limited spectrum.

Hyperspectral imaging was employed in the study of Liu et al. [18] to determine physical stress in humans and investigate the changes in oxygen saturation of facial tissue over time. Individual differences exist in the alterations in facial StO₂ during physical stress. It was discovered that the five ROIs—the forehead, left cheek, right cheek, nose and median cleft—were useful in differentiating between physical stress and a relaxed condition. SVM classifier achieved the best performance with an accuracy of 82.11%. The study only measured facial StO₂, which is one of many physiological indicators of stress. It is possible that combining facial StO₂ with other physiological measures would improve the accuracy of stress identification. The study used a relatively simple classification algorithm (SVM). More complex algorithms, such as DL, may be able to achieve even higher accuracy.

D. Using EDA Signals

To assess physiological reactions, Sandulescu et al. [19] used a wearable device. The participant's non-dominant hand was fitted with the wearable device, which recorded electrodermal activity (EDA) and pulse plethysmograph (PPG) signals on both fingers. With a software program, heart rate variability (HRV) and PPG autocorrelation signal were also retrieved. When employing SVM to classify an individual as stressed or not, an accuracy of 82% was attained.

Electrodermal activity (EDA) signals can be broken down into phasic (fast-changing) and tonic (slow-changing) components, according to Greco et al. [20]. Researchers can determine more about an individual's mental state by computing attributes derived from these components. Convex optimization can also be used to derive the phasic component's sparse sudomotor nerve activity (SMNA) driver.

E. Using EEG Signals

A ML (ML) framework was presented by Subhani et al. [21] for the use of electroencephalogram (EEG) signals to identify stress at various levels. The scientists induced four stress levels in 22 volunteers using an experimental paradigm, and then they employed EEG data analysis to determine the subjects' stress levels. The maximum accuracy that the suggested framework was able to attain was 94.6% between two stress levels and the correlation between stress levels was 83.43%. To confirm the results of this investigation, more research is required and

offer a robust ML system for the multiple level detection of stress in larger and more diverse populations.

Schmidt et al. [22] employed three-class emotion classification and binary stress identification as two tests to assess the performance of deep neural networks. 15 participants in baseline, amused, and stressed conditions provided the same datasets, which were used. Traditional ML techniques were surpassed by deep neural networks, as demonstrated by the outcomes of both tournaments. The deep neural networks outperformed the LDA ML technique, which yielded an 92.83% (accuracy rate) and 91.07% (F1 score), for binary stress recognition, with 99.80% (accuracy rate) and 99.67% (F1 score). Random Forest ML produced 95.34% (accuracy rate) and of 94.78% (F1 score) for 3-class emotion categorization, compared to 98.12% for deep neural network. This study's utilization of physiological data from a restricted set of sensors is one of its limitations. Perhaps more sensors or other forms of data—like facial expressions or EEG recordings—could be used to enhance the deep neural networks' capabilities. Lastly, the study did not assess how resilient deep brain networks were to various stressor kinds or stress levels. It is important to determine how well the networks can detect stress in a variety of situations before they can be deployed in real-world settings.

F. Using Hybrid Model

A hybrid model (HB) that blends gradient boosting machine (GBM) and random forest (RF), two tree-based ensemble models, was proposed by Ding et al. [23]. They demonstrated that HB outperforms individual models or linear models by achieving 100% accuracy on a limited dataset. On short datasets for stress identification, tree-based models outperform linear based models in terms of accuracy. Performance can be enhanced by combining models rather than using individual models. While DL models can be computationally expensive, they might be useful for huge datasets. When the features in the dataset have a correlation with the target classes, ML models can attain a high degree of accuracy. Because of the tiny dataset employed in the study, the conclusions cannot be broadly applied. The research did not assess the effectiveness of HB in comparison to other DL models for stress identification. The study did not explore how HB could be used to develop a real-world stress identification system.

G. Using various other approaches

For studies pertaining to stress and user modeling, Koldijk et al. [24] developed a brand-new multimodal dataset known as the SWELL-KW dataset. These tasks, which included writing, reading, searching, and other common knowledge tasks, were given to 25 people in order to collect the dataset. Time limits and email interruptions were two stressors that were employed to control the participants' working environment. Among the recorded data were information on skin conductance, heart rate, facial expressions, bodily postures, and computer logging. This publicly available dataset included both preprocessed and raw data with features extracted. Validated quizzes on workload, cognitive

effort and related issues were used to evaluate the working behavior.

A neural network with a deep sparse networking was presented by Lin et al. [25] as an automated technique for stress detection utilizing cross-media microblog data. Using data of microblog posts from people who had received a stress diagnosis, the authors assessed the proposed methodology. They found that, with an accuracy rate of 82.29%, their approach performed better than others. The suggested strategy benefited from the use of the language and visual components of the microblog records. This indicates that the use of many data types can improve the accuracy of stress identification. A dataset of microblog postings from individuals diagnosed with stress was used to evaluate the proposed method. It's unclear whether the technique can be used to a bigger population. A complex deep learning model, which can be costly to train and compute to implement, was employed in the recommended method.

An additional investigation on the identification of the work environment was conducted by Ceja et al. [26]. Through the use of data from cellphone accelerometer sensors, he found activity linked to the individuals' stress levels. This particular sensor was chosen because it has fewer privacy concerns than location, audio, or video recording. The low power consumption of this sensor is one of the reasons it was chosen for incorporation into wearables such as fitness trackers, which are smaller in size. Thirty people in two distinct groups were given smartphones. By employing user-specific models and comparable users, they were able to attain accuracy rates of 60% and 71%, respectively. They could only use one smartphone's accelerometer data, which is not sufficient for recognition of stress.

A variety of techniques for examining physiological data collected from body-attached sensors were described by Sioni et al. [27]. The main aim is to identify stress and categorize emotions according to these indicators. Physiological computing solutions for stress identification have the potential to benefit users, notwithstanding the limitations of existing sensors. He stressed how important it is to find solutions for these problems and develop novel strategies in order to improve accuracy and comfort.

Five intensity stages and eight different states were differentiated using machine learning (ML) in a temperament identification system created by Zenonos et al. [28]. Using subject-independent accuracy, the system scored 62.14%.

Tzirakis et al. [29] developed a deep neural network model to detect stress in videos.

Song et al. [30] developed an effective stress categorization model using a Deep Belief Network (DBN) by analyzing data from the sixth Korea National Health and Nutrition Examination Survey (2013-2015). Using statistical analysis, nine important stress detection characteristics were found. With an ideal setup of one hidden layer and six nodes, their DBN model performed better than other ML models. The model's accuracy and specificity were 66.23% and 75.32%.

al. [31] introduced a unique technique called Multi-channel Data Synchronization (MCDS) for non-contact physiological signal (PS) identification. The algorithm is composed of four primary steps: PS/baseline classification, StO2 signal synchronization, face region of interest (ROI) selection, and StO2 extraction. The StO2 extraction algorithm, which is based on the Beer-Lambert (BL) model, was also thoroughly explained by the authors. In 88.3% of scenarios, the authors' algorithm can correctly identify PS. The program can only subjectively differentiate between PS and baseline states; it cannot statistically measure stress levels. Further analysis is necessary because the study was conducted in a lab setting, making it complex to evaluate the algorithm's efficacy in real-world situations.

By applying bidirectional LSTM with attention and post-training via remote supervision on stress tweets, Winata et al. [32] developed a model for categorizing stress that was improved upon based on interview transcriptions. With 74.1% accuracy and 74.3% f-score, the model finds important phrases that indicate stress, such as "employment," "pressure," and "anxious." Negation and other intricate grammatical constructions are among the semantic concepts it fails to capture. A more thorough system for stress recognition should include prosodic characteristics and contextual information, rather than being restricted to transcriptions of spoken language.

Facial gestures are another important indicator of stress levels in people. Marias et al. [33] also developed a system for recognizing and understanding emotional states of tension based on facial gestures recorded in video. The features under investigation were head action parameters, mouth-related actions, eye-related actions, and photoplethysmographic heart rate estimate based on camera. The participants were to be seated 50 centimeters apart in front of a computer screen with an integrated camera. Techniques like the AdaBoost classifier, Naive Bayes classifier, SVM, CNN, and Generalized Likelihood Ratio were used and assessed. During the social exposure experiment, the Adaboost classifier gave the best classification results, with an accuracy of 91.68%.

Since mental stress is a significant issue that might have disastrous effects, Ahuja et al. [34] investigated it in college students, particularly during examinations. He made use of the Perceived Stress Scale (PSS) exam. The Perceived Stress Scale (PSS) is a self-reported quiz that takes into account a person's assessment of how unexpected, uncontrollable, and overwhelming life has been over the last month. It is a commonly used instrument for measuring stress levels in both clinical and scientific contexts. The Support Vector Machine method produced an accuracy of 85.71%. A self-reported stress indicator with potential bias is the PSS test. To increase the prediction models' accuracy, more objective stress signals, such as physiological markers, could be employed.

Deep convolutional neural networks(DCCN) were employed by Li et al. [35] to achieve accuracy rates of 99.80% and 99.55% for binary and three-class classification, respectively. The accuracy rates of the deep multilayer perceptron neural network for binary and three-class categorization were 99.65% and 98.38%, respectively. Comparing the networks' performance to previous techniques that examined physiological data for binary stress detection and three-class emotion categorization, there was a noticeable improvement. Future studies aiming to expand on this experiment will need to train and evaluate the two neural networks on considerably larger datasets containing a variety of human groups. Because the networks would be exposed to a more realistic picture of the global population, this would strengthen their robustness. 15 human participants were involved to prepare the dataset, which may not adequately represent the whole human population in general.

The ethical considerations and data privacy issues in stress identification research necessitate a careful approach. Obtaining informed consent, anonymizing or de-identifying personal data, and ensuring robust data security are paramount. Transparency, data minimization, and purpose limitation are essential to build trust with participants. Researchers must provide user control over data, seek ethical review board approval, and address biases to ensure fairness. Clear policies on long-term data storage and public education about the study's ethical safeguards are crucial for responsible and respectful research practices. It is also important to note that cultural and demographic factors influence stress responses, potentially introducing biases in DL and ML models used for stress identification. In AI-driven mental health research, obtaining informed consent is critical to respect individuals' autonomy and ensure transparency. Robust data security measures must be implemented to protect sensitive information, addressing ethical concerns and fostering trust in the responsible use of technology for mental health applications. In AI-driven mental health research, obtaining informed consent is critical to respect individuals' autonomy and ensure transparency. Robust data security measures must be implemented to protect sensitive information, addressing ethical concerns and fostering trust in the responsible use of technology for mental health applications.

III. CONCLUSION

Numerous modalities, such as electromyography, ECG signals, hyperspectral imaging, and EEG signals, have demonstrated potential in the research on stress recognition using DL and ML techniques. Notable advancements involve the integration of deep learning frameworks, outperforming conventional methods in accuracy. Challenges persist, such as the need for larger, diverse datasets, validation in real-world stressors, and the development of non-intrusive monitoring systems. Multimodal data fusion is suggested for improved accuracy, though generalization and model robustness are ongoing concerns. Evaluating physiological signals comprehensively, acknowledging self-report limitations, is crucial. Future research should focus on refining model resilience, addressing privacy concerns, and developing real-world applications. Incorporating diverse data sources, expanding datasets, exploring real-world applications, and testing across various stressors will enhance robustness and practicality. Enhancing the interpretability and explainability of deep learning models

should be done to make the decision-making process transparent, fostering trust among users and stakeholders. Secure and ethical data frameworks, advanced algorithms like reinforcement learning, and collaborations between researchers and industry stakeholders are vital. Investigating individual differences, including cultural nuances and personality traits, can lead to personalized solutions. Mechanisms should be developed for incorporating user feedback into stress recognition models, allowing individuals to actively participate in refining and personalizing their stress management tools. Overall, future research aims for a holistic understanding of stress, leveraging technology for accessible, reliable, and adaptable real-world stress identification tools.

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